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$$\Rightarrow x + y + z = 1$$

$$ax + y + az = 1$$

$$x + ay + z = 1$$

$$ax + y + z = 1$$

$$\begin{bmatrix} 1 & 1 & 1 & | & 1 \\ a & 1 & a & | & 1 \\ 1 & a & 1 & | & 1 \\ a & 1 & 1 & | & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 0 & 1-a & 0 & | & 1-a \\ 0 & a-1 & 0 & | & 0 \\ 0 & 1-a & 1-a & | & 1-a \end{bmatrix} \sim$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 0 & 1-a & 0 & | & 1-a \\ 0 & 0 & 0 & | & a-1 \\ 0 & 0 & 1-a & | & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 0 & 1-a & 0 & | & 1-a \\ 0 & 0 & 1-a & | & 0 \\ 0 & 0 & 0 & | & a-1 \end{bmatrix}$$

$a \neq 1 \rightarrow$ syst. ni rešljiv

$$a = 1:$$

$$\begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$x + y + z = 1$$

$$x = 1 - (y + z)$$

$$\vec{r}(y, z) = (1, 0, 0) + y(-1, 1, 0) + z(-1, 0, 1)$$

Determinanta

$$\det A = \sum_{\sigma \in \text{Sym}(n)} \text{sgn}(\sigma) A_{1\sigma(1)} A_{2\sigma(2)} \dots A_{n\sigma(n)}$$

$\underbrace{\hspace{10em}}_{n! \text{ členov} \rightarrow \text{ťažko}}$

①

Izračunaj det. po definiciji:

$$A = \begin{bmatrix} 9 & 7 & 6 & 8 & 5 \\ 3 & 0 & 0 & 2 & 0 \\ 5 & 3 & 0 & 4 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 7 & 5 & 4 & 6 & 0 \end{bmatrix}$$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 2 & 1 & 3 \end{pmatrix}$$

$$\text{sgn } \sigma = 1$$

$$\begin{array}{l} (54) (52) (51) (53) \\ (42) (41) (43) \\ (21) \end{array}$$

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$$\det A = 5 \cdot 2 \cdot 3 \cdot 1 \cdot 4 = 5! = 120$$

② 2 Gaussove metode izračunaj det A

$$A = \begin{bmatrix} 213 & 344 & 186 & 162 & 137 \\ 344 & 157 & 295 & 106 & 89 \\ 419 & 418 & 419 & 418 & 416 \\ 417 & 416 & 417 & 416 & 416 \end{bmatrix} \sim \begin{bmatrix} 51 & 49 & 25 & 137 \\ 49 & 51 & 89 & 106 \\ 0 & 0 & 1 & 418 \\ 0 & 0 & 1 & 416 \end{bmatrix} \sim$$

$$\begin{array}{ccccccc} 213 & 344 & 186 & 157 & 162 & 295 & 89 \\ -162 & -295 & -137 & -106 & -137 & -106 & -25 \\ \hline 51 & 49 & 49 & 51 & 25 & 89 & 64 \end{array}$$

$$\sim \begin{bmatrix} 2 & -2 & -64 & 31 \\ 49 & 51 & 89 & 106 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 1 & 416 \end{bmatrix} \sim \begin{bmatrix} 2 & 0 & -64 & 31 \\ 49 & 100 & 89 & 106 \\ 0 & 0 & 1 & 416 \\ 0 & 0 & 0 & 2 \end{bmatrix} \sim$$

$$\sim \begin{bmatrix} 2 & 0 & -64 & 31 \\ 49 & 100 & 89 & 106 \\ 0 & 0 & 1 & 416 \\ 0 & 0 & 0 & 2 \end{bmatrix} \sim \begin{bmatrix} 100 & 49 & 89 & 106 \\ 0 & 2 & -64 & 31 \\ 0 & 0 & 1 & 416 \\ 0 & 0 & 0 & 2 \end{bmatrix} \rightarrow \det A' = 400$$

menjava vrstic $\rightarrow \det \rightarrow -\det$

$v_i \rightarrow \alpha v_i \rightarrow \det \rightarrow \alpha \det$

$$\det A = -400$$

3. Izračunaj $\det A$ z razvojem po vrstici/stolpcu

$$A_n = \begin{pmatrix} 5 & 2 & 0 & \dots & 0 \\ 2 & 5 & 2 & \dots & 0 \\ 0 & 2 & 5 & \dots & 2 \\ \vdots & & & \ddots & \vdots \\ 0 & \dots & 0 & 2 & 5 \end{pmatrix} \leftarrow$$

$$\det A_n = 5 \cdot \det \begin{pmatrix} 5 & 2 & & \\ 2 & 5 & & \\ & & \ddots & \\ & & & 5 & 2 \\ & & & 2 & 5 \end{pmatrix}^{(n-1) \times (n-1)} -$$

$$- 2 \cdot \det \begin{pmatrix} 2 & 2 & 0 & \dots \\ 0 & 5 & 2 & \dots \\ 0 & 2 & 5 & 2 & \dots \\ \vdots & & & \ddots & \\ & & & & 5 & 2 \\ & & & & 2 & 5 \end{pmatrix} =$$

$$= 5 \cdot \det A_{n-1} - 2 \cdot 2 \det A_{n-2} = 5 \det A_{n-1} - 4 \det A_{n-2}$$

$$\det A_n = D_n$$

$$D_n = 5D_{n-1} - 4D_{n-2}$$

$$p^2 - 5p + 4 = 0$$

$$(p-4)(p-1) = 0$$

$$p_1 = 4$$

$$p_2 = 1$$

$$D_1 = 5$$

$$D_2 = \begin{vmatrix} 5 & 2 \\ 2 & 5 \end{vmatrix} = 21$$

$$D_n = A_1 \cdot 4^n + A_2$$

$$5 = A_1 \cdot 4 + A_2$$

$$21 = A_1 \cdot 16 + A_2$$

$$A_2 = 5 - 4A_1$$

$$21 = 16A_1 + 5 - 4A_1$$

$$A_2 = 5 - \frac{16}{3} = -\frac{1}{3}$$

$$16 = 12A_1$$

$$A_1 = \frac{4}{3}$$

$$D_n = \frac{4}{3} 4^n - \frac{1}{3} = \frac{1}{3} (4^{n+1} - 1)$$